INCH-POUND

A-A-52411(AT)
October 9, 1992

COMMERCIAL ITEM DESCRIPTION

INSULATION, MOLDED, NOISE BARRIER

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-I-62520(AT), which is canceled.

1.0 Abstract. This commercial item description (CID) covers an expandable, self-skinning insulation material intended for use in applications where a chemical-resistant, sealed, molded, noise barrier is required.

2.0 Salient characteristics.

2.1 Materials. The insulation shall be as specified herein. The use of recovered material made in compliance with section 402 of the Clean Air Act (33 U.S.C. 1342 et. seq.) is acceptable providing that all requirements of this CID are met.

2.2 Design and configuration. The design and configuration of the insulation shall be as specified in the applicable drawing (see 5.2).

2.2.1 Color. The color of the insulation shall be black.

2.3 Physical properties.

2.3.1 Surface density. The surface density of 1-inch thick insulation shall be 1.2 pounds per square foot ± 10 percent.

2.4 Performance.

2.4.1 Tear resistance. The tear resistance of the insulation shall be not less than 4 pounds force per inch (ppi) when tested in accordance with ASTM D624, die C.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000.

AMSC N/A FSC 5640

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2.4.2 Tensile strength. The tensile strength of the insulation shall be not less than 25 pounds per square inch (psi) when the insulation is heat aged at 250 ± l0°F for 48 hours and tested in accordance with ASTM D412, die C.

2.4.3 Elongation. The elongation of the insulation when heat aged at 250 ± l0°F for 48 hours shall be not less than 80 percent when tested in accordance with ASTM D412, die C.

2.4.4 Flammability. The burn rate of the insulation shall be not greater than 2 inches per minute when tested in accordance with SAE J369.

2.4.5 Impact resistance. When using a Gardner impact tester having a 1-kilogram dart with a 1.6 centimeter diameter, insulation 2 inches by 4 inches by 1-inch thick, shall be conditioned at 77 ± 5°F and 50 ± 5% relative humidity for 7 days and then conditioned at -50 ± 2°F for not less than 4 hours. While in this conditioned state the insulation shall be capable of withstanding an impact of not less than 15 centimeter-kilograms. After being allowed to return to room temperature for 15 minutes, the insulation shall be bent around a 2-inch radius bar in the tested area and examined for cracks or other evidence of failure.

2.4.6 Flexibility. Insulation, 1 inch by 8 inches by 1-inch thick, shall be conditioned at 77 ± 5°F and 50 ± 5% relative humidity for 7 days; 250°F for 4 hours with a 1-hour recovery at 77 ± 5°F; and -50°F for 4 hours with a 1-hour recovery at 77 ± 5°F. While in this conditioned state, the insulation shall be capable of being bent around a 1-inch mandrel without cracking.

2.4.7 Acoustical performance. The acoustical performance of 1-inch thick insulation mounted on 20-gauge steel shall be as specified in table I and tested in accordance with ASTM E90.

TABLE I. Noise barrier properties.

Frequency (hertz)	Sound transmission loss (decibels)
160	16
250	20
500	21
1000	36
2000	50
4000	58

2.5 Material safety data sheet (MSDS). An MSDS shall be prepared in accordance with FED-STD-313 (see 5.2 and 5.6).

2.6 Identification and marking. Unless otherwise specified in the drawing, identification and marking shall be permanent and legible and shall include as a minimum the manufacturer's name, the part number, and the national stock number (NSN). Identification and marking shall be accomplished without adversely affecting the acceptance and function of the finished product.

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2.7 Workmanship. Workmanship shall be such that the insulation is uniform in color, mechanical properties and density, and is free of defects which would adversely affect its life, serviceability, or appearance.

3.0 Quality assurance provisions.

3.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

3.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.0 Preservation, packaging packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see 5.2).

5.0 Notes.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

5.1 Addresses for obtaining copies of referenced documents.

5.1.1 Government documents. FED-STD-313 "Materials Safety Data, Transportation Data and Disposal Data for Hazardous Materials" is available from the Navy Publications and Printing Service office, Standardization Documents order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

5.1.2 Non-Government publications. ASTM D412 “Standard Test Methods for Rubber Properties in Tension"; ASTM D624 “Standard Test Methods for Rubber Property-Tear Resistance"; E90 "Standard Test Method for Laboratory Measurements of Airborne Sound Transmission Loss of Building Partitions"; and ASTM E380 “Standard and Practices for Use of the International System (SI) are available from the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, PA 19103. SAE J369 "Flammability of Polymetric Interior Materials" is available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of the Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation and the specific issue of individual documents referenced (see 5.1).
c. Title, number, and revision letter of the applicable engineering drawing and part number (see 2.2 and 2.6).
d. Identify activities requiring copies of completed MSDS and the number of copies required (see 2.5 and 5.5).
e. Applicable levels of preservation, packaging, packing, labeling and marking (see 4).

5.3 Cross-reference data. Insulation conforming to this CID is interchangeable/ substitutable with insulation conforming to MIL-I-62520(AT).

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5.4 Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

5.5 MSDS. The contracting officer should identify those activities requiring copies of the completed MSDS prepared in accordance with FED-STD-313. Additional pertinent Government mailing addresses for submission of data are listed in appendix B of FED-STD-313 (see 2.5 and 5.2).

5.6 Metric product. Insulation manufactured to metric dimensions will be considered on the following basis:

a. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met.

b. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

c. The contracting officer has the option of accepting or rejecting the product.

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